

Geometrical tolerances: the 14 ISO 1101 symbols and how to read them

1. The 14 ISO 1101 symbols and their tolerance zone

The tolerance zone is the space in which the actual feature must lie; t is the value written in the frame, in millimetres. When the value is preceded by Ø, the zone is cylindrical (or circular).

Symbol	Characteristic	Family	Datum	Tolerance zone
—	Straightness	Form	No	Two parallel lines t apart; a cylinder of diameter t when the tolerance (preceded by Ø) applies to an axis
▭	Flatness	Form	No	Two parallel planes t apart
○	Roundness	Form	No	In each cross-section, two concentric circles whose radii differ by t
⊘	Cylindricity	Form	No	Two coaxial cylinders whose radii differ by t
⌒	Line profile	Form, orientation or location	Optional	Two lines enveloping circles of diameter t centred on the theoretically exact profile
⌒	Surface profile	Form, orientation or location	Optional	Two surfaces enveloping spheres of diameter t centred on the theoretically exact surface
	Parallelism	Orientation	Yes	Two planes parallel to the datum, t apart; a cylinder Ø t parallel to the datum for an axis
⊥	Perpendicularity	Orientation	Yes	Two planes perpendicular to the datum, t apart; a cylinder Ø t perpendicular to it for an axis
∠	Angularity	Orientation	Yes	Two planes t apart, inclined at the theoretically exact angle to the datum
⊕	Position	Location	Yes, usually	Zone centred on the theoretically exact position: cylinder Ø t for an axis, two planes t apart for a median plane
⊙	Concentricity / coaxiality	Location	Yes	Circle Ø t (concentricity of a point) or cylinder Ø t (coaxiality of an axis) centred on the datum
≡	Symmetry	Location	Yes	Two planes t apart, symmetrical about the datum median plane
↗	Circular run-out	Run-out	Yes (axis)	In each section, two circles concentric to the datum axis, t apart (radial); two circles on a measuring cylinder (axial)
↗	Total run-out	Run-out	Yes (axis)	Two cylinders coaxial with the datum axis, t apart (radial); two planes perpendicular to the axis (axial)

These tolerances are independent of size tolerances (ISO 8015 independency principle): a $\varnothing 20 \pm 0.1$ dimension limits neither roundness nor axis straightness, unless the envelope requirement E applies. Unstated form and orientation tolerances fall under the title-block general tolerance: see classes H, K, L in ISO 2768 general tolerances.

3. Modifiers

Symbol	Name	Effect
Ⓜ	Maximum material requirement	The tolerance grows as the feature departs from its maximum material condition (larger hole, smaller shaft): the "bonus" that guarantees assembly at the lowest cost.

Symbol	Name	Effect
Ⓛ	Least material requirement	The mirror of Ⓜ: used to guarantee a minimum wall thickness.
ⓔ	Envelope requirement	Written after a size tolerance (Ø 20 H7 ⓔ), not in the frame: the part must not violate the perfect-form envelope at maximum material. Essential on fits.
Ⓟ	Projected tolerance zone	The zone is moved outside the part, over the stated height: typical for a tapped hole receiving a stud or a bolt passing through another part.
ⓕ	Free state	The tolerance applies to the unrestrained part (flexible parts, seals, thin sheet).
CZ	Common zone	One zone for several separate features, for example two faces that must be coplanar.

Reference values taken from the article <https://www.baouch.fr/en/blog/tolerances-geometriques-symboles-iso-1101/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.